

The Arena Sakharov Presupposed: How Event Density Supplies the Medium Induced Gravity Needs

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Preamble — what this article does and does not claim

1. **This grounds the two things Sakharov's induced gravity *presupposes*, not a from-nothing derivation.** Sakharov (1967; modern review Visser, gr-qc/0204062) showed that the one-loop effective action of matter fields on a **given** Lorentzian manifold automatically contains the cosmological constant, the Einstein-Hilbert action, and curvature-squared terms. Two things are assumed and left to a deeper theory: the **manifold** itself (“assume you are given a Lorentzian manifold”) and the **cutoff** that regulates the divergent vacuum sum. Event Density (ED) supplies both. This article arranges ED's standing results against Visser's account; it does not re-derive Sakharov's loop.
2. **ED supplies the arena at *acoustic-metric* tier.** ED derives an emergent Lorentzian metric ($g \sim 1/b$, Paper_GR-I) from its substrate — a genuine “reasonably large, reasonably flat arena,” and with a *real medium* underneath it (the thing analog-gravity models lack). But this is the kinematic/acoustic metric; the **full curvature-emergence program** (deriving the metric's curvature dynamics and its coupling coefficients from the raw participation graph) is the standing **open** bridge under every geometric result in the corpus, and is not closed here.
3. **ED does not carry out the one-loop induction of R on its own manifold.** Sakharov's mechanism (the heat-kernel / Seeley-DeWitt expansion of the matter determinant) is *his*; ED's emergent matter would realize it on ED's emergent arena, but that computation is not done here. What ED contributes is the arena and the cutoff the mechanism requires.
4. **ED dissolves Sakharov's cosmological-constant *catastrophe*, inheriting only Λ 's value.** Sakharov's *continuum* one-loop $\Lambda \sim \kappa^4$ overshoots the observed value by $\sim 10^{122}$. ED's substrate is *discrete*, so there is no infinite mode-tower and that $\sim M_P^4$ term is structurally absent (the report's “dissolved catastrophe”; Paper_038_5; UV-finiteness Paper_040, Paper_111) — ED does **not** share the catastrophe. It identifies Λ 's form ($\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$) and reframes its smallness as the Friedmann ratio $(H_0/M_P)^2$, with only the *value* (via H_0 , Route-A open) inherited. So on the cosmological constant, ED does *better* than Sakharov's continuum induction, not worse; the residual is deriving H_0 , not curing an overshoot.
5. **ED's G is substrate-native, not Sakharov's spectrum formula.** ED's metric stiffness comes from the substrate's own always-on connectivity, a *different source* than integrating out a matter loop, so ED does not reproduce Sakharov's “ G^{-1} from the particle spectrum $\text{str}[k_1]$.” ED gives an induced-*strength* G from the grain; Sakharov's matter loop would be an additional contribution on top.

6. **ED's results are conditional derivations from thirteen posited primitives**, never derivations from nothing, and the framework is not experimentally confirmed. Tiers are marked per claim.
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1. What Event Density is, and the family this belongs to

Event Density is a discrete, relational substrate: a participation graph whose channels carry a bandwidth $b \geq 0$ and a phase, with one process primitive, **commitment** (P11), the irreversible act that writes the arrow of time into the law. Gravity is emergent: the metric is the *reading* of the bandwidth field's connectivity ($g \sim 1/b$, Paper_GR-I), and Newton's constant is a conversion factor tied to the grain ($G = c^3 \ell_P^2 / \hbar$, the constants ledger, Paper_027).

This is the fourth paper in the *microscopic-completions* family, each of which takes a respected “gravity from a deeper principle” result and shows ED supplies the microscopic input its authors flagged as owed to a deeper theory: Jacobson's equation of state (Paper_GravityAsEquationOfState), Padmanabhan's horizon equipartition (Paper_GravityAsHorizonEquipartition), the khronometric Clausius derivation (Paper_KhronometricEquationOfState), and now Sakharov's induced gravity. The through-line is the same asset in different clothes: a Planck-grain, participation-based substrate beneath a presupposed geometry.

2. Sakharov's move, and the two things he presupposes

Following Visser: Einstein gravity has two ingredients — (1) the pseudo-Riemannian geometry (from the equivalence principle) and (2) the field equations for the Ricci tensor. **Sakharov addresses only (2)**. He assumes you are handed a Lorentzian manifold, makes no assumption about its dynamics, does *not* quantize the geometry, and quantizes everything else. Then the one-loop matter effective action is *guaranteed* to contain

$$\int d^4x \sqrt{-g} \{c_0 + c_1 R(g) + c_2 (R^2)\},$$

which is, term for term, the cosmological constant, the Einstein-Hilbert action $R/16\pi G$, and higher-curvature corrections. Using the heat-kernel expansion with a short-distance cutoff κ (dimensionally $[\kappa] = \kappa^{-2}$), the divergences organize by power:

- a κ^4 divergence induces the cosmological constant, $\Lambda \sim \kappa^4$;
- a κ^2 divergence induces Newton's constant, $\frac{1}{G} \approx -\frac{1}{2\pi} \mathrm{str}[k_1]$, (Visser eq. 29), so **G^{-1} scales as the cutoff squared, weighted by the particle spectrum** through the Seeley-DeWitt coefficient k_1 ;
- logarithmic divergences induce the curvature-squared couplings.

The central lesson Sakharov draws, and the one this article builds on, is **not** the particle-spectrum formula but the deeper structural fact Visser states in his conclusion:

“Deriving the inverse square law, and all of Einstein gravity, is in fact automatic once you have demonstrated the existence of Lorentzian manifolds — reasonably large reasonably flat arenas on which to set up a low-energy QFT.”

In other words, Sakharov **reduces the problem of gravity's dynamics to the problem of the arena's existence**. Given a manifold, the dynamics comes for free at one loop. The two things left owed to a

deeper theory are therefore precisely: **the manifold** (Sakharov assumes it) and **the cutoff** (Sakharov leaves it to Planck-scale physics). ED supplies both.

3. ED supplies the arena — from a real substrate, not by assumption

Sakharov’s manifold “flaps in the breeze”: it is posited, an external classical background with no account of where it came from. ED’s is not posited. The emergent metric is the *reading* of the substrate’s bandwidth connectivity:

- **The metric form is derived.** Paper_GR-I derives the emergent line element with $g_{ij} \sim b^{-1}$, lapse $N^2 \sim b$, and the Schwarzschild relation $g_{00} g_{rr} = -1$ on the Einstein branch. This is a genuine emergent **Lorentzian** geometry — a “reasonably large, reasonably flat arena” for a low-energy effective theory, exactly what Sakharov’s construction needs.
- **And it has a real medium.** Where analog-gravity models supply an acoustic metric but hedge it as an *analogy* (there is no claimed real medium under real spacetime), ED supplies the medium: the discrete participation substrate. This is the ontological upgrade Sakharov’s presupposed manifold lacks. ED’s arena is not “assume a manifold”; it is “the manifold is what the substrate’s connectivity looks like when coarse-grained.” (That ED’s emergent metric *is* an acoustic metric with a real medium under it is the thesis of the analog-gravity completion, target #4 in this folder’s README; it is invoked here as the ontological upgrade to Sakharov’s arena, and established there, not here.)

Honest tier. The metric is derived at *weakfield* (GR-I, khronometric, coinciding with general relativity there), which is precisely the tier Sakharov’s *local* construction actually uses — a locally-flat patch to host a QFT. The metric *form* and the gravitational effective-action *form* (khronometric, one open stiffness coefficient) are derived; deriving that coefficient and the full $\sqrt{-g}$, R dynamics from the raw graph is the open **curvature-emergence** program. The corpus establishes the (ii)-grade emergent metric as achievable and an (i)-grade first-principles derivation of the full curved Einstein dynamics as *not* achievable under current axioms (ED_Geometry_Emergence_Scoping), and flags curvature emergence open under every geometric result (Paper_AreaLawIsTheEdgeCount). So ED supplies the arena Sakharov needs at the tier his local construction uses, while the global, fully-dynamical curvature account remains ED’s open frontier.

4. ED supplies the cutoff — and more: a substrate-native rigidity of gravitational strength

Sakharov’s induced $G^{-1} \sim \kappa^2$ makes the load-bearing input a bare cutoff κ , handed to Planck-scale physics. ED **has** that cutoff natively: the grain $\ell_{ED} = \ell_P$ (P08), $\kappa = 1/\ell_P$. Substituting gives ED’s constants-ledger value

$$G = \frac{c^3 \ell_P^2}{\hbar},$$

the “first shadow” — a conversion factor at the seam where mass meets geometry, absent from the entire Standard Model, which the ledger reads as a tell that gravity is not fundamental (Constants_Ledger_and_G_Dimension, Paper_027). The dimensional content matches Sakharov exactly, $G^{-1} \sim \kappa^2 \sim 1/\ell_P^2$ — though this is a *cutoff identification* (ED’s grain = Sakharov’s κ), not a derivation of G ’s numerical value, which is inherited (Paper_027).

But ED supplies *more* than the cutoff, and here is the one genuinely new observation of this paper, stated carefully. ED already **derives** a metric stiffness independent of any matter loop:

$$M_P^2 \sim \frac{s_{02}}{\ell_P^2},$$

the coefficient of the spatial-curvature energy, set by the **always-on** P02 reciprocal edge-sharing of the substrate — Planck-dense, present at every adjacency every tick whether or not anything commits (Paper_GR-IV §4). This is dimensionally and conceptually Sakharov’s “metric elasticity” $G^{-1} \sim \kappa^2$, but its **source is different**: it is the substrate’s own standing connectivity, not the vacuum fluctuations of matter integrated out at one loop. So:

In Sakharov’s picture, gravity’s rigidity is **induced by matter** (no matter loop, no G). In ED, gravity’s rigidity is the substrate’s **own standing elasticity** — pre-matter, kinematic — and Sakharov’s matter loop would be an *additional* contribution riding on top. ED does not need the particle spectrum to have gravity; it needs the grain and the substrate’s sharing.

This is a real distinction, not a re-labeling: ED’s metric rigidity is a property of the substrate’s standing connectivity, so ED obtains a G of the right *strength* ($\sim \kappa^2 \sim 1/\ell_P^2$) from the grain and the sharing fraction alone, without needing a matter spectrum to generate it. Note the terminology: ED’s rigidity is *not* “induced” in Sakharov’s sense — his term of art for a coupling generated by the matter loop — precisely because it is standing / always-on rather than loop-generated. Sakharov’s matter-loop contribution would then *add* to this substrate-native rigidity rather than being its sole source. (This is a claim about the *source* of the rigidity, not about its spatial variation: like Newton’s constant in either picture, ED’s M_P^2 is essentially constant, set by the grain, not tracking local matter density.)

5. The synthesis, and its honest boundary

Put the pieces together with Sakharov’s own reduction. Sakharov: *given the arena, the dynamics is automatic*. ED: *here is the arena (emergent metric, real medium) and the cutoff (grain), and moreover a substrate-native rigidity*. The two views compose:

- ED builds the arena the substrate coarse-grains into (§3);
- ED supplies the cutoff and a derived induced-strength G (§4);
- Sakharov’s one-loop mechanism — which ED’s *emergent* matter would realize on that arena — then generates the *form* of the Einstein-Hilbert action automatically (alongside the induced Λ and R^2 terms), as it does on any manifold.

And this dovetails with the rest of the family: ED already reaches the Einstein equation two other independent ways — its dynamical bandwidth rule (Paper_GR-III) and Jacobson’s clean-equilibrium equation of state (Paper_GravityAsEquationOfState, Paper_KhronometricEquationOfState). Sakharov adds a third lens: the same Einstein dynamics is what the matter loop induces on ED’s emergent arena. (The induced-gravity program even has its own horizon-entropy branch — Frolov-Fursaev’s induced-gravity black-hole entropy — that sits directly alongside ED’s Jacobson/Padmanabhan horizon results; the microscopic degrees of freedom that induced gravity attributes to heavy matter fields would plausibly correspond, in ED, to the severed participation channels — an analogy worth flagging, not a computation done here.)

The boundary, stated plainly. What this paper does *not* deliver:

- the one-loop induction of R computed on ED's *own* emergent manifold with ED's *own* emergent-matter spectrum (that is a real calculation, not attempted; Sakharov's result guarantees its *form*, not its coefficients in ED);
- the closed curvature-emergence bridge (the open coefficient of ED's gravitational effective action; §3);
- a derivation of Λ 's *value*. But — correcting a natural first reading — ED does **not** reproduce Sakharov's *catastrophe*: his continuum one-loop $\Lambda \sim \kappa^4$ genuinely overshoots by $\sim 10^{122}$, whereas ED's *discrete* substrate has no infinite mode-tower to sum, so that $\sim M_P^4$ term is structurally absent (the report's “dissolved catastrophe”; Paper_038_5; the huge number “was the size of the mistake, not a real quantity waiting to be cancelled”). ED identifies Λ 's form ($\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$, tied to the cosmic horizon; Paper_KM-II) and reframes its smallness as the Friedmann ratio $(H_0/M_P)^2 \approx 10^{-122}$, two ordinary scales rather than a fine-tuning. Only Λ 's *value* (via H_0) is inherited; the residual hierarchy (deriving H_0 , Route-A) is open. So **induced gravity's worst problem — the catastrophe — is dissolved by ED's discreteness; only the last measured scale is inherited**. (The naive grain/V1-cutoff estimate in Paper_038_5 §3.4 that overshoots by $\sim 10^{60}$ is the *failed* estimate the discreteness supersedes, not ED's position.)

6. Honest tiers

- **Derived (ED)**: the emergent metric form $g \sim 1/b$ (Paper_GR-I, conditional/khronometric); the substrate-native metric stiffness $M_P^2 \sim s_{\{02\}}/\ell_P^2$ (Paper_GR-IV §4); $G = c^3\ell_P^2/\hbar$ (constants ledger).
- **Grounded / assembled (literature + ED)**: the identification of ED's grain with Sakharov's cutoff; the reading of ED's arena as the manifold Sakharov presupposes; the synthesis of §5. Sakharov's one-loop result is Visser's, cited, not re-derived.
- **Acoustic-tier / open**: ED's arena is the kinematic/acoustic metric; full curvature emergence (coefficients from the graph) is open (ED_Geometry_Emergence_Scoping, Paper_AreaLawIsTheEdgeCount).
- **Not attempted**: the one-loop induction of R on ED's emergent manifold with ED's emergent matter.
- **Dissolved + value-inherited**: the cosmological-constant *catastrophe* is dissolved (discrete substrate, no infinite tower); Λ 's form is identified and its smallness reframed as the Friedmann ratio $(H_0/M_P)^2$; only Λ 's *value* (via H_0) is inherited (Paper_038_5, the report's “dissolved catastrophe”). The value of the grain is inherited.

7. Position statement

Sakharov's induced gravity is often read as “gravity comes from the vacuum fluctuations of matter.” Visser's sharper reading is that Sakharov *reduced* gravity to the existence of an arena: given a Lorentzian manifold to host a quantum field theory, all of Einstein gravity is automatic at one loop, and the only things owed to a deeper theory are the manifold and the cutoff. Event Density is a candidate for that deeper theory, and it supplies exactly those two owed things: it **builds** the arena as the coarse-grained reading of a discrete participation substrate — an emergent Lorentzian metric with, for once, a *real medium* beneath it — and it **is** the cutoff, the Planck grain that fixes $G = c^3\ell_P^2/\hbar$. It even supplies a substrate-native gravitational rigidity of the right strength ($\sim \kappa^2$), sourced by the substrate's standing connectivity rather than by a matter loop, so that ED does not wait on a particle spectrum to set G . What ED does not yet do is compute the induced action on its own arena, close its own curvature-emergence bridge, or derive Λ 's value — though on

the cosmological constant it dissolves the catastrophe (its discreteness leaves no infinite tower to overshoot) and inherits only the value via H_0 , rather than sharing Sakharov’s embarrassment. The claim is therefore precise and bounded: **induced gravity presupposes a medium and a cutoff; Event Density supplies both, and gives the medium a real substrate — the arena Sakharov assumed, now built rather than borrowed.** None of this is offered as confirmed physics; it is offered as a grounding of the inputs Sakharov’s construction was waiting for.

Cross-references

- Sakharov, A. D., *Sov. Phys. Dokl.* **12**, 1040 (1968); Visser, M., “Sakharov’s induced gravity: a modern perspective,” *Mod. Phys. Lett. A* **17**, 977 (2002), gr-qc/0204062 (read in full: the one-loop effective action eqs. 1–2; $1/G \sim \text{str}[k_1]\kappa^2$ eq. 29; the “reasonably large reasonably flat arenas” conclusion, §9).
- **ED arena / curvature emergence:** physics-papers/gravity/Paper_GR-I_WeakFieldEinsteinMetric.md ($g \sim 1/b$); event-density/theory/ED_Geometry_Emergence_Scoping.md (§3F Sakharov, “presupposes a metric”; (ii)-grade achievable, (i)-grade not under current axioms); physics-papers/substrate-evaluation/Paper_AreaLawIsTheEdgeCount.md (curvature emergence flagged open).
- **ED cutoff / stiffness / G:** physics-papers/gravity/Paper_GR-IV_ArrowsAlibi.md §4 ($M_P^2 \sim s_{\{0\}}/\ell_P^2$, always-on P02 sharing); physics-papers/constants-inherited/Constants_Ledger_and_G_Dimension.md + Paper_027_Newtons_G.md ($G = c^3\ell_P^2/\hbar$, the first shadow).
- **ED Λ (the inherited problem):** physics-papers/gravity/Paper_038_5_Lambda_V1_Cosmological.md ($\sim 10^{60}$ overshoot, on the record); Paper_KM-II_KhrononCosmology.md §7 ($\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$).
- **Family companions:** Paper_GravityAsEquationOfState.md (Jacobson), Paper_GravityAsHorizonEquipartition.md (Padmanabhan), Paper_KhronometricEquationOfState.md (khronometric Clausius); README.md.
- **Induced-gravity horizon entropy (adjacent):** Frolov–Fursaev, e.g. hep-th/9703178, gr-qc/9907046 (black-hole entropy in Sakharov’s induced gravity).